Chapter 9. Bootstrap Tests

Parametric and nonparametric bootstrap tests, duality with confidence intervals

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The problem

A test needs the distribution of the statistic *under* H_0, but the bootstrap mimics the *actual* F, which may not satisfy H_0. There are two ways around this.

1. **Resample under the null.** Build an estimate $\hat{F}_0$ that satisfies H_0, and resample from it.
2. **Resample a pivot.** If $T_\theta = (\hat{\theta} - \theta)/\widehat{\text{se}}$ is (approximately) pivotal, its law is the same under H_0 and under the truth. Approximate it by $T^* = (\hat{\theta}^* - \hat{\theta})/\widehat{\text{se}}^*$ resampled from $\hat{F}$, and compare with the observed T_{θ_0}.

In both cases the p-value is $\hat{p} = \{1 + \#(T^{*b} \text{ at least as extreme as } T_{\text{obs}})\}/(B+1)$.

9.1 Parametric bootstrap tests

In a model $F(\cdot; \theta, \eta)$ with nuisance η, test $H_0 : \theta = \theta_0$ with a statistic $S(\mathbb{X}, \theta_0)$ that is large under H_1.

- *Null resampling:* compute the restricted maximum likelihood estimate $\hat{\eta}_0$ under $\theta = \theta_0$, draw samples from $F(\cdot; \theta_0, \hat{\eta}_0)$, and recompute $S(\mathbb{X}^*, \theta_0)$.
- *Pivot resampling:* draw from $F(\cdot; \hat{\theta}, \hat{\eta})$ and recompute $S(\mathbb{X}^*, \hat{\theta})$.

Proposition 9.1

For $X_i \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$ and $S = \sqrt{n}(\bar{X} - \mu_0)/s$, both schemes reproduce the exact null distribution t_{n-1}.

Proof. Under null resampling the bootstrap sample is $N(\mu_0, \hat{\sigma}_0^2)$, so $\sqrt{n}(\bar{X}^* - \mu_0)/s^* \sim t_{n-1}$ whatever $\hat{\sigma}_0$ is. Under pivot resampling the sample is $N(\hat{\mu}, \hat{\sigma}^2)$ and $\sqrt{n}(\bar{X}^* - \hat{\mu})/s^* \sim t_{n-1}$. ■

Exact pivots are not always available. For Laplace(μ, b), $|X_i - \mu|/b \sim \text{Exp}(1)$, so $2\sum_i |X_i - \mu|/b \sim \chi^2_{2n}$ is a pivot for μ only if b is known. With b unknown one falls back on approximate pivots (e.g. the CLT-based t-statistic), or on null resampling from Laplace$(\mu_0, \hat{b}_0)$ with $\hat{b}_0 = \frac{1}{n}\sum |X_i - \mu_0|$. With two samples and a common scale, the null model fits a common center (the pooled median) and a pooled $\hat{b}_0$.

In general the two schemes agree asymptotically. Studentized statistics give more accurate tests than raw differences (Hall and Wilson). A non-pivotal statistic, such as a ratio $\hat{\theta}/\theta_0$ whose spread depends on θ, should only be used with null resampling.

9.2 Nonparametric bootstrap tests

One sample, location. For $H_0 : \theta = \theta_0$ in $X_i = \theta + \epsilon_i$:

- null resampling: resample from the recentered data $\tilde{X}_i = X_i - \hat{\theta} + \theta_0$, which satisfy H_0;
- pivot resampling: resample from the raw data and use $T^* = (\hat{\theta}^* - \hat{\theta})/\widehat{\text{se}}^*$.

For a mean and the t-statistic the two are identical: recentering shifts every bootstrap mean by the same constant. A two-sided test uses $|T^*| \geq |T_{\text{obs}}|$.

Two samples, $H_0 : F = G$. Pool the samples and draw both bootstrap samples from the pooled empirical distribution. This is the with-replacement analogue of the permutation test, and the two give similar answers except in very small samples.

Two samples, equal locations with possibly different shapes. Here $F \neq G$ is allowed under $H_0 : \mu_1 = \mu_2$, so pooling is wrong and permutation tests are invalid. Use pivot resampling within each sample:

$$T^* = \frac{\bar{Y}^* - \bar{X}^* - (\bar{Y} - \bar{X})}{\sqrt{s_X^{*2}/m + s_Y^{*2}/n}}, \qquad T_{\text{obs}} = \frac{\bar{Y} - \bar{X}}{\sqrt{s_X^2/m + s_Y^2/n}}.$$

Equivalently, recenter each sample to the common pooled mean and resample within samples. Centering both samples at a common value and then using the *unstandardized* difference without subtracting $\bar{Y} - \bar{X}$ does not give a null distribution; it reproduces the sampling distribution of the observed difference.

9.3 Duality with bootstrap intervals

Proposition 9.2

The two-sided level-α pivot test of $H_0 : \theta = \theta_0$ based on $\hat{\theta} - \theta$ (resp. on $(\hat{\theta} - \theta)/\widehat{\text{se}}$) accepts exactly when θ_0 lies in the basic (resp. bootstrap-t) interval.

Proof. The test accepts iff $Q_{H^*}(\frac{\alpha}{2}) \leq \hat{\theta} - \theta_0 \leq Q_{H^*}(1 - \frac{\alpha}{2})$, i.e. iff $\hat{\theta} - Q_{H^*}(1 - \frac{\alpha}{2}) \leq \theta_0 \leq \hat{\theta} - Q_{H^*}(\frac{\alpha}{2})$, which is (8.1). The studentized case gives (8.2) in the same way. ■

Percentile-type intervals have no simple test counterpart, because the transformation φ behind them is unknown.